



GE Medical Systems

Technical Publications

Direction 2333495
Revision 0

Signa® ReFlex AP Option Installation for MGD

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Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Immediately complete a "Damage Loss Claim Form", available via MS Exchange Mail, after the damage is found.

MS Exchange Path:

Outlook/Public Folder/All Public Folders/Medical Systems/!Global Initiatives/Information Management/Forms/Common Forms/DAMAGE LOSS CLAIM FORM.

Send the completed form to the email address listed in the form.

For more information about the Transportation Claim Procedure, access the GE Medical Systems Intranet and enter the following URL address (case sensitive):

<ftp://3.87.40.2/globepro/qualsys/Docs/190016MF.PDF>

Rev. 11/15/2000

Direction 2128126 - Language Policy For Service Documentation, Rev 0

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTES AVISOS PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・このサービスマニュアルは英語版しかありません。
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- ・このサービスマニュアルを熟読し、理解せずに装置のサービスを行わないでください。
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注意:

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

<u>REV</u>	<u>DATE</u>	<u>PRIMARY REASON FOR CHANGE</u>
A.....	Apr 2, 2002	Preliminary version.
0.....	Apr 29, 2002	Initial release.

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* This revision/letter corresponds to the indicated document's revision control system.

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This option is only compatible for systems with an MGD version System Cabinet.

1- INTRODUCTION

This document covers new installations and upgrades of the ReFlex100, ReFlex200, and ReFlex400 AP options for MGD.

Total installation time for the hardware is up to one hour for one installer (depending on option being installed).

1-1 Signa 10.x Options Installed by this Manual

- M3000RD – EXCITE ReFlex 100
- M3000RE – EXCITE ReFlex 200
- M3000RF – EXCITE ReFlex 400
- M3090EF – EXCITE ReFlex 100 to 200 Upgrade Kit
- M3090EG – EXCITE ReFlex 100 to 400 Upgrade Kit
- M3090EH – EXCITE ReFlex 200 to 400 Upgrade Kit

1-2 Contents

M3000RD (ReFlex100):

- **One** motherboard populated with **one** daughter card
- Rating Plate

M3000RE (ReFlex200):

- **One** motherboard populated with **two** daughter cards
- Rating Plate

M3000RF (ReFlex400):

- **One** motherboard, jumper-configured as a master AP
- **One** motherboard, jumper-configured as a slave AP
- Each mother board will be populated with **two** daughter cards
- **One** double-slot cable and cover kit
- Rating Plate

M3090EF (ReFlex100 to 200):

- **One** daughter card
- Rating Plate

M3090EG (ReFlex100 to 400):

- **One** motherboard populated with **two** daughter cards
- **One** daughter card
- **One** double-slot cable and cover kit
- Rating Plate

M3090EH (ReFlex200 to 400):

- **One** motherboard populated with **two** daughter cards
- **One** double-slot cable and cover kit
- Rating Plate

2- REFLEX AP INSTALLATION (NEW INSTALLATIONS)

This section covers catalogs M3000RD, M3000RE, and M3000RF. The MGD chassis can be configured with either one or two AP motherboards. Each motherboard can be configured with one or two processor daughtercards. Each daughtercard has two G4 processors.

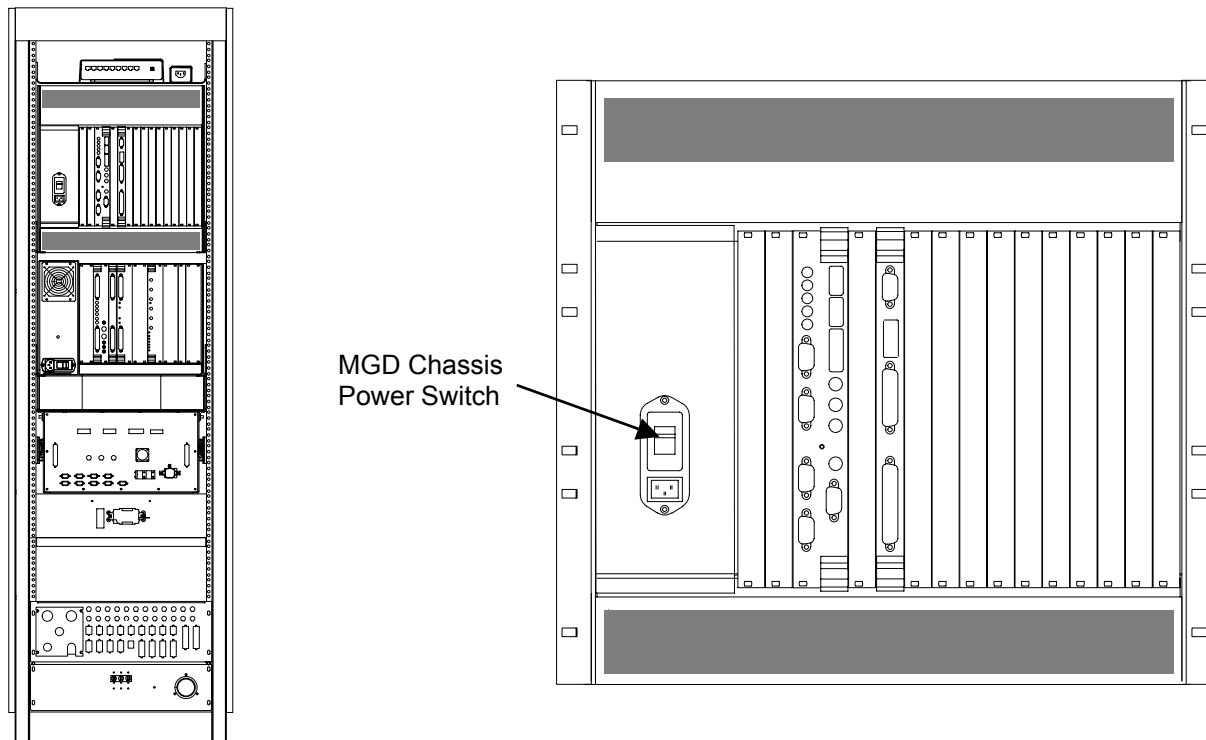
Mercury's "hardware" name of the motherboard is MCPJ6.



EQUIPMENT DAMAGE POSSIBILITY. Electro Static Discharge (ESD) precautions must be used when removing the AP assembly or replacing any modules on the AP.

2-1 Power Off Procedure

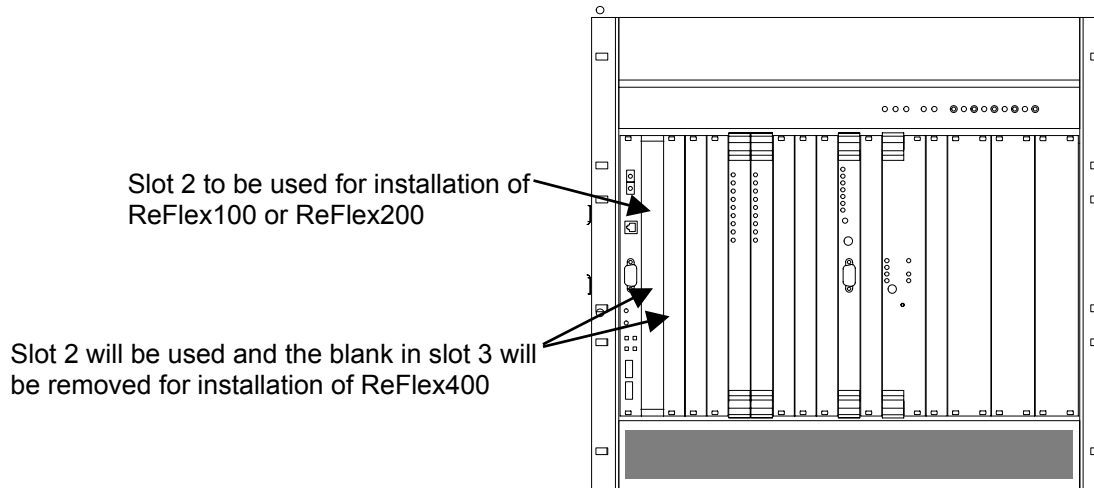
1. Remove the System Cabinet front cover and open rear door.
2. Turn off power switch at rear of MGD chassis. See Illustration 2-1.



SYSTEM CABINET POWER OFF LOCATIONS
ILLUSTRATION 2-1

2-2 Installation of Option

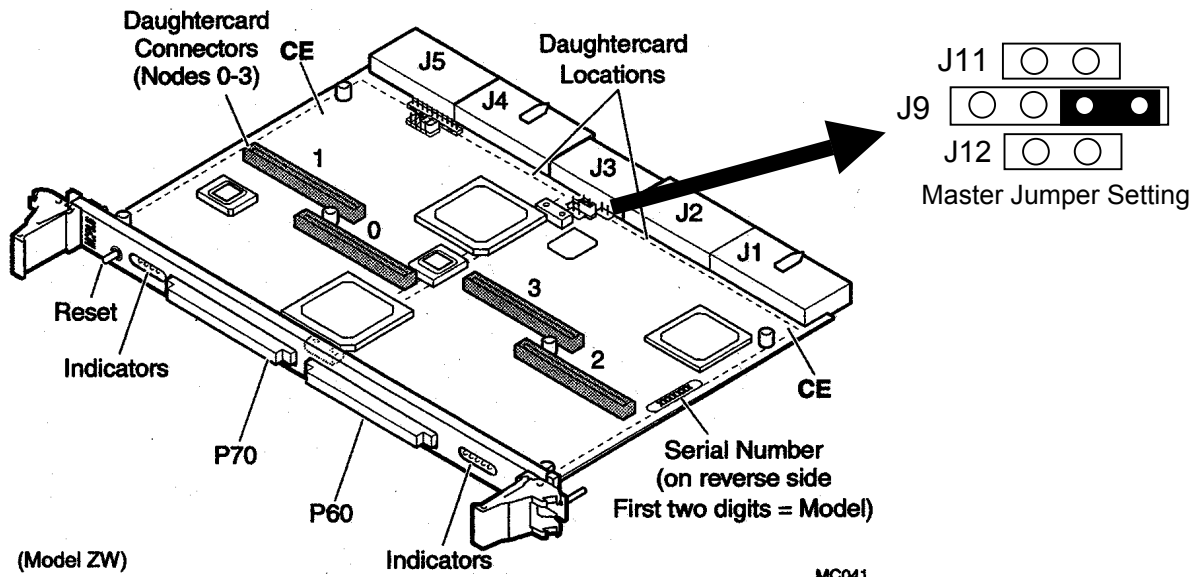
If M3000RD or M3000RE are being installed, then only slot 2 on the front side of the MGD Chassis will be used. If M3000RF is being installed, slot 2 will be used and blank on slot 3 will need to be removed. See Illustration 2-2.



REFLEX BOARD LOCATIONS
ILLUSTRATION 2-2

2-2-1 Installation of ReFlex100 and 200

The ReFlex100 will have one daughtercard. The ReFlex200 will have two daughtercards. For either, make sure the board is jumpered as a MASTER. See Illustration 2-3.



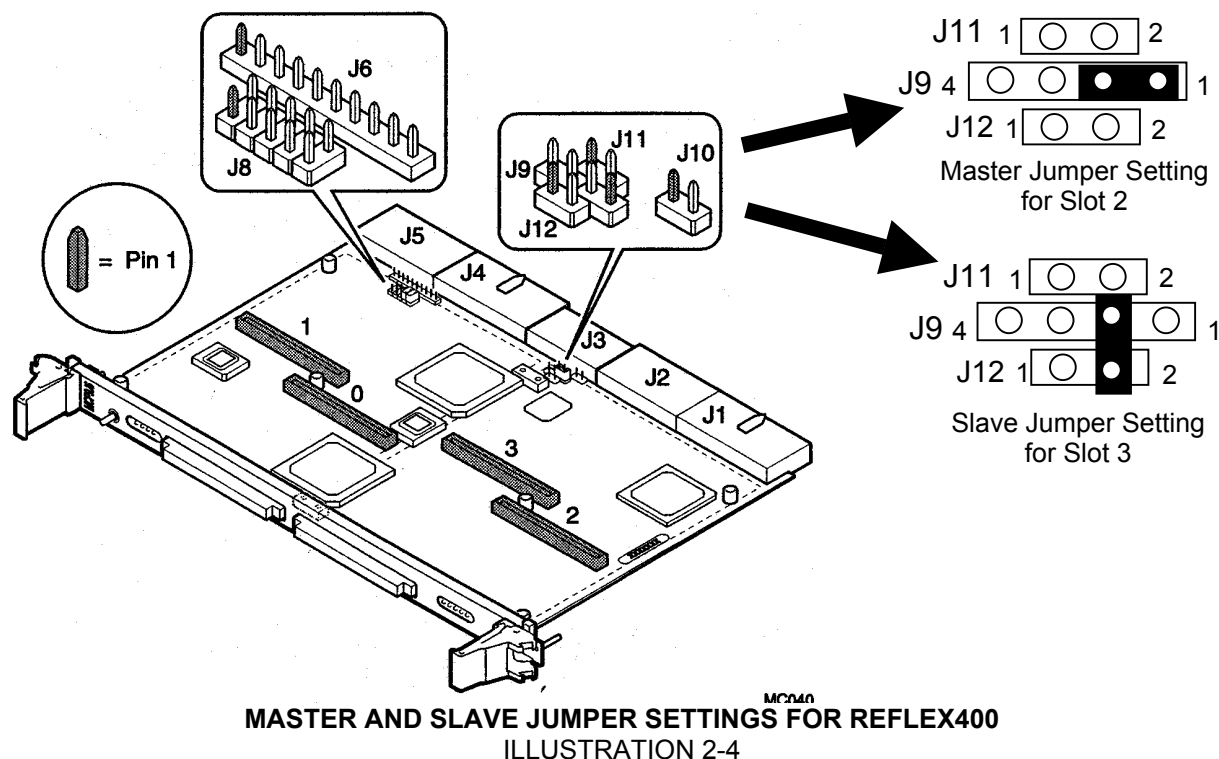
REFLEX BOARD WITH MASTER JUMPER SETTING
ILLUSTRATION 2-3

1. Carefully insert board into slot 2.
2. Fasten in place with screws at top and bottom of front plate.

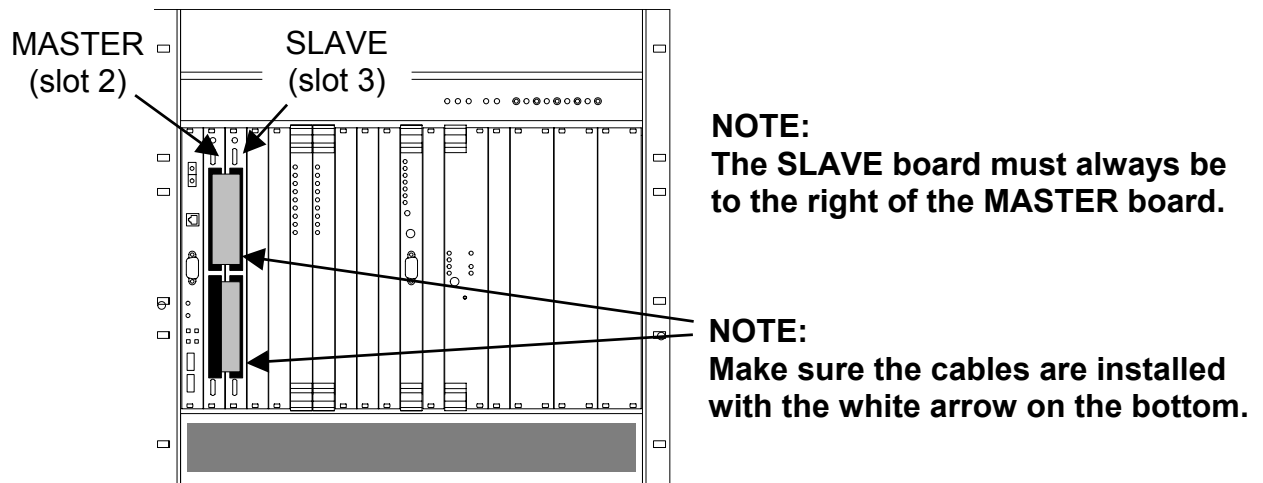
2-2-2 Installation of ReFlex400

The ReFlex400 is made up of two boards with two daughtercards on each board. Each board will require a jumper setting. Refer to Illustration 2-4 for jumper settings.

1. The board inserted into slot 2 must be set as the MASTER RACEway clock generator with jumper on pins 1 and 2 on J9.
2. The board inserted into slot 3 will be the SLAVE and must be set to receive the clock synch signal over the front panel P70 connector. The jumper must be placed on pin 2 on J12 and pin 2 on J9.



3. Carefully insert MASTER board into slot 2 and SLAVE board into slot 3. See Illustration 2-5.
4. Fasten in place with screws at top and bottom of front plate.



INSTALLATION OF REFLEX400 IN MGD CHASSIS
ILLUSTRATION 2-5

1. Install the two RACEway adapter cables to the front panel ports, P60 and P70. There is a small white arrow on one end of each cable, and each cable must be attached such that the arrow is on the bottom. See Illustration 2-5.
2. As a double check that both cables are in the proper orientation, each cable should be attached such that the end with the white arrow is connected to the SLAVE board.
3. Attach covers over adapter cables.

2-3 Power On and TPS Reset

After installing the motherboard(s) and attaching the RACEway Adapter cables (only for the dual board configuration), power on the MGD chassis, then do a **[TPS Reset]**.

There is no need to modify the configuration file because the system software will auto-configure the new board(s).

2-4 Functional Check

To verify the new installed hardware was configured by the system, use the **Hardware Tracker (HART)** software tool as follows:

1. On the host computer, click the Service Desktop icon and (if not already open) start the MR Service Desktop by clicking the **[Service Browser]** button on the left side of the screen.
2. Select the "Error Logs" tab.
3. Select "HART logs".
4. View the HART Current Status log file to verify the new installed hardware is present in the MGD board location(s) it was installed in.

3- REFLEX AP UPGRADE

This section covers M3090EF, M3090EG, and M3090EH. The ReFlex100 AP can be upgraded to a ReFlex200 with the addition of a daughtercard or upgraded to a ReFlex400 with the addition of daughtercard and motherboard with two daughtercards.

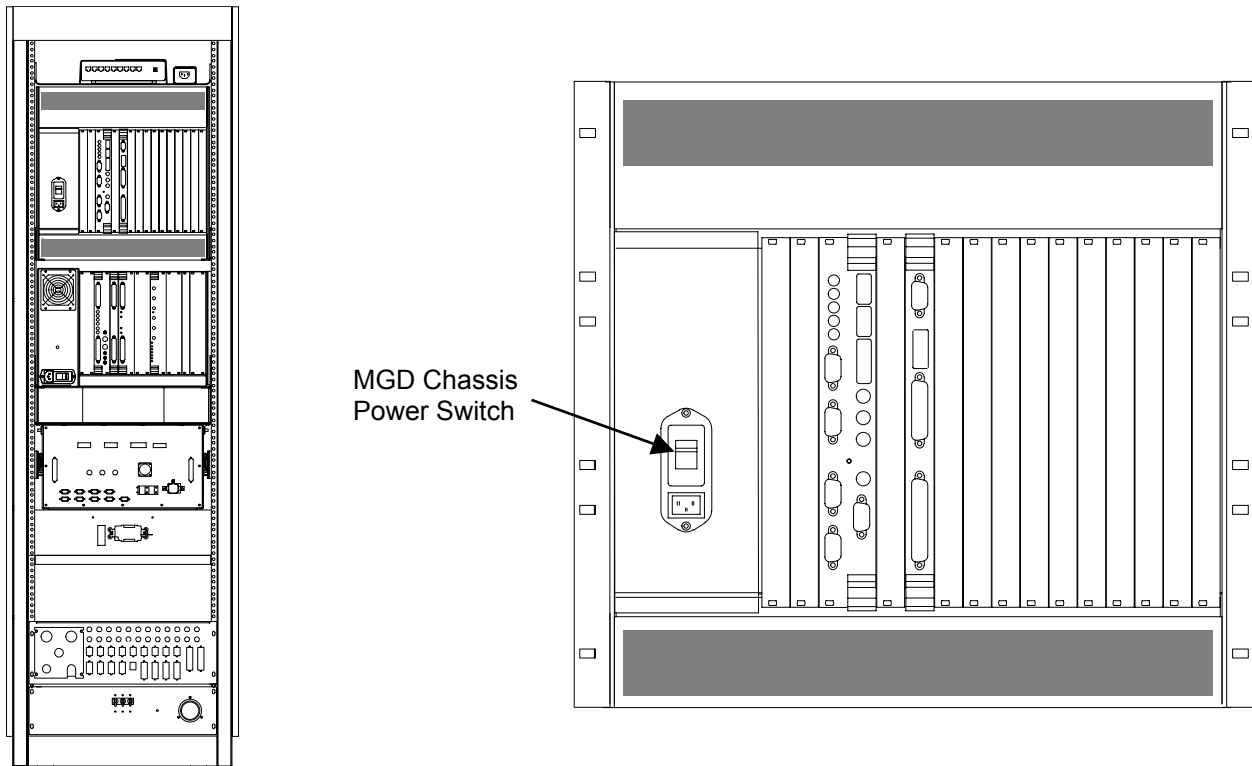
The ReFlex200 can be upgraded to a ReFlex400 with the addition of a motherboard with two daughtercards.



EQUIPMENT DAMAGE POSSIBILITY. Electro Static Discharge (ESD) precautions must be used when removing the AP assembly or replacing any modules on the AP.

3-1 Power Off Procedure

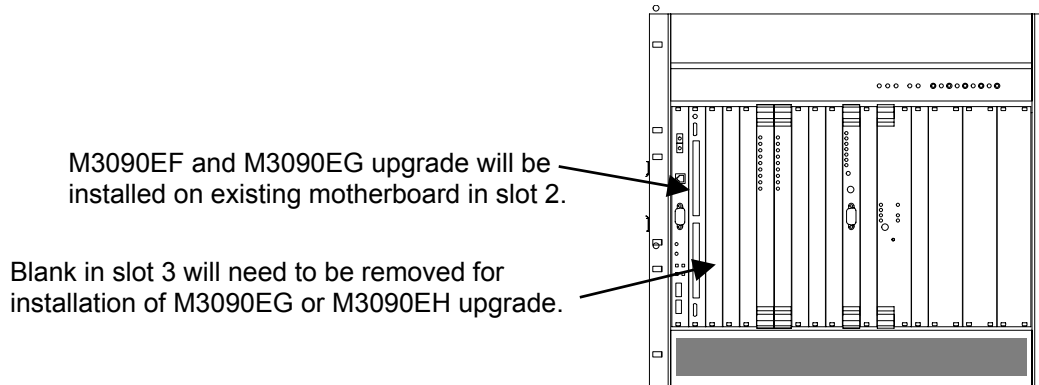
1. Remove the System Cabinet front cover and open rear door.
2. Turn off power switch at rear of MGD chassis. See Illustration 3-1.



POWER OFF LOCATIONS
ILLUSTRATION 3-1

3-2 Installation of Option

If the M3090EG or M3090EH upgrade is being installed, then only the blank panel on slot 3 on the front side of the MGD Chassis will need to be removed. If the M3090EF upgrade is being installed, then the blank panel on slot 3 will not be removed. See Illustration 3-2.

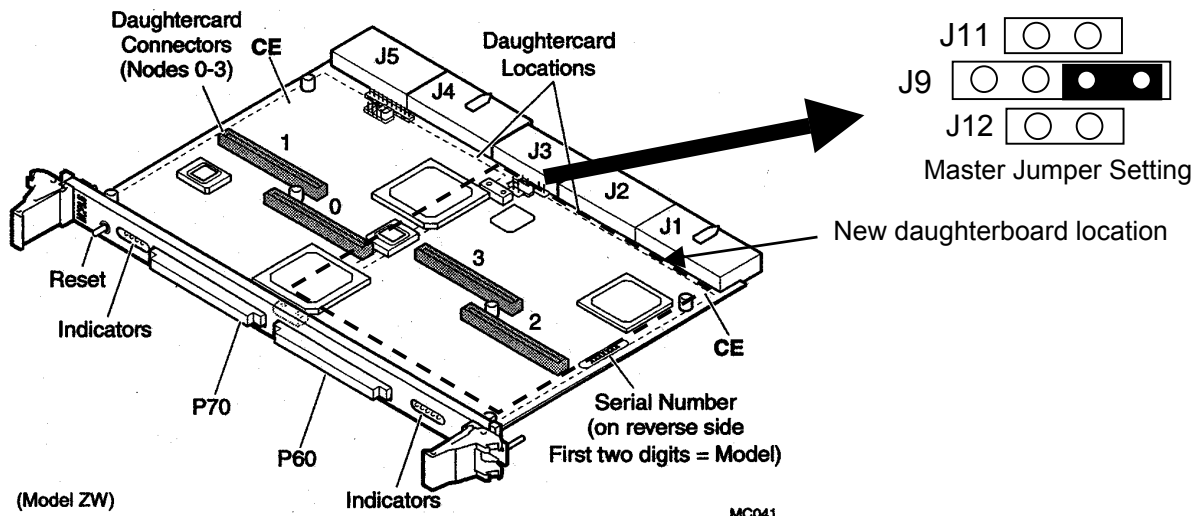


REFLEX UPGRADE LOCATIONS
ILLUSTRATION 3-2

3-2-1 Installation of ReFlex100 to ReFlex200 Upgrade, M3090EF

The ReFlex200 upgrade will add one daughtercard to the existing motherboard in slot 2.

1. Loosen the two front panel screws in slot 2. Using appropriate anti-static precautions, remove the AP Assembly from the MGD Chassis. See Illustration 3-2.



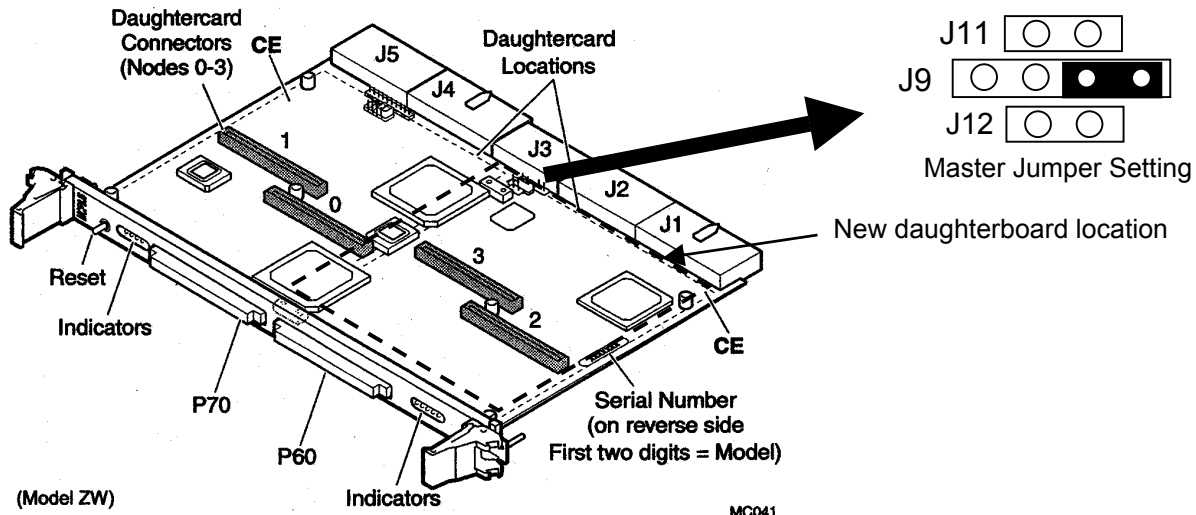
DAUGHTERBOARD LOCATION
ILLUSTRATION 3-3

2. On the front panel, remove the single slot EMI covers at P60 and P70.
3. Carefully install new daughterboard to motherboard and attach with four screws. See Illustration 3-3.
4. Carefully insert the board back into slot 2.
5. Fasten in place with screws at top and bottom of front plate.

3-2-2 Installation of ReFlex100 to ReFlex400 Upgrade, M3090EG

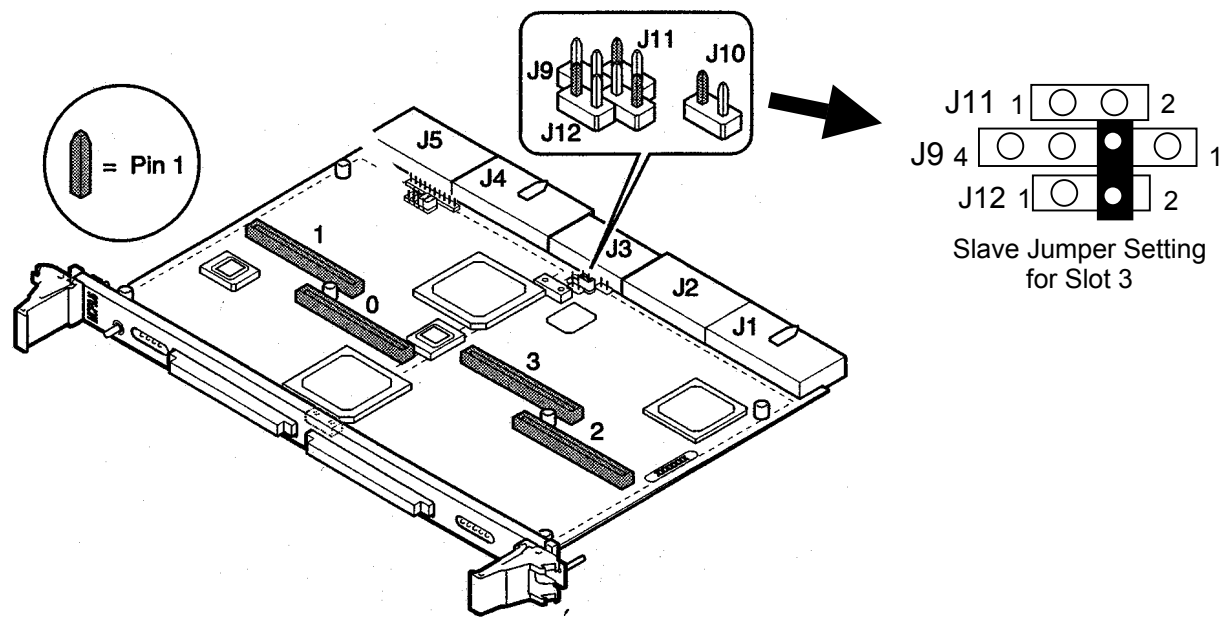
This ReFlex400 upgrade will add one daughtercard to the existing motherboard in slot 2 and add a new motherboard with 2 daughterboards to be installed in slot 3.

1. Loosen the two front panel screws in slot 2. Using appropriate anti-static precautions, remove the AP Assembly from the MGD Chassis (see Illustration 3-2).



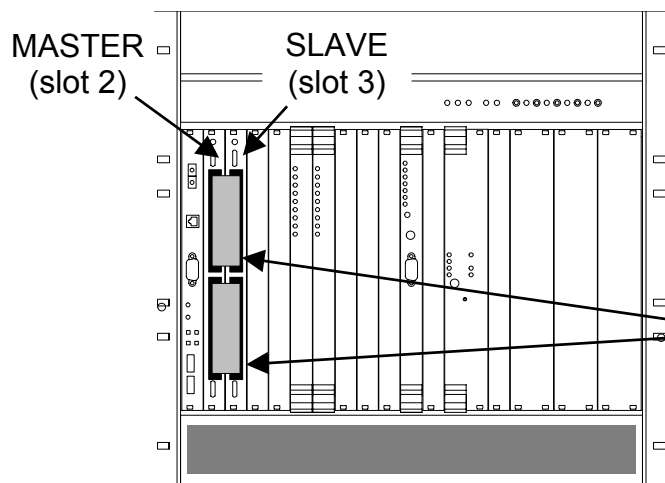
DAUGHTERBOARD LOCATION
ILLUSTRATION 3-4

2. On the front panel, remove the single slot EMI covers at P60 and P70.
3. Carefully install new daughterboard to motherboard and attach with four screws. Make sure board jumper is set as the MASTER RACEway clock generator with jumper on pins 1 and 2 on J9. See illustration 3-4.
4. Carefully insert the board back into slot 2.
5. Fasten in place with screws at top and bottom of front plate.
6. Remove blank panel in slot 3.
7. Remove new Motherboard with 2 daughterboards attached from packaging. Make sure the jumper is set to SLAVE so it can receive the clock synch signal over the front panel P70 connector. The jumper must be placed on pin 2 on J12 and pin 2 on J9. See Illustration 3-5.



SLAVE JUMPER SETTINGS FOR REFLEX400 IN SLOT 3
ILLUSTRATION 3-5

8. Carefully insert board into slot 3.
9. Fasten in place with screws at top and bottom of front plate.
10. Install the two RACEway adapter cables to the front panel ports, P60 and P70. There is a small white arrow on one end of each cable, and each cable must be attached such that the arrow is on the bottom. See Illustration 3-6.
11. As a double check that both cables are in the proper orientation, each cable should be attached such that the end with the white arrow is connected to the SLAVE board.
12. Attach covers over adapter cables.



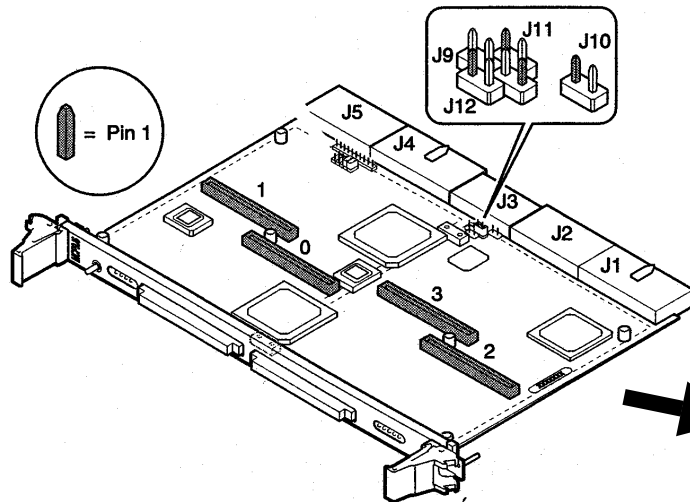
- NOTE:**
The SLAVE board must always be to the right of the MASTER board.
- NOTE:**
Make sure the cables are installed with the white arrow on the bottom.

INSTALLATION OF REFLEX400 IN MGD CHASSIS
ILLUSTRATION 3-6

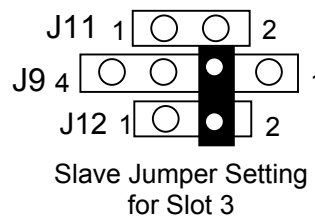
3-2-3 Installation of ReFlex200 to ReFlex400 Upgrade, M3090EH

This ReFlex400 upgrade will add a new motherboard with 2 daughterboards to be installed in slot 3.

1. Remove blank panel in slot 3.



2. Remove new Motherboard with 2 daughterboards attached from packaging. Make sure the jumper is set to SLAVE so it can receive the clock synch signal over the front panel P70 connector. The jumper must be placed on pin 2 on J12 and pin 2 on J9. See Illustration 3-7.



SLAVE JUMPER SETTINGS FOR REFLEX400 IN SLOT 3
ILLUSTRATION 3-7

3. Carefully insert board into slot 3 and fasten in place with screws at top and bottom of front plate.
4. Install the two RACEway adapter cables to the front panel ports, P60 and P70. There is a small white arrow on one end of each cable, and each cable must be attached such that the arrow is on the bottom. See Illustration 3-6.
5. As a double check that both cables are in the proper orientation, each cable should be attached such that the end with the white arrow is connected to the SLAVE board.
6. Attach covers over adapter cables.

3-3 Power On and TPS Reset

After installing the motherboard(s) and attaching the RACEway Adapter cables (only for the dual board configuration), power on the MGD chassis, then do a **[TPS Reset]**.

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3-4 Functional Check

To verify the new installed hardware was configured by the system, use the **Hardware Tracker** (HART) software tool as follows:

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3. View the HART Current Status log file to verify the new installed hardware is present in the MGD board location(s) it was installed in.